

Large multimodal model-driven clinical decision support systems: mapping architectures to decision requirements — a scoping review protocol

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1. Introduction

1.1. Background

The current landscape of Large Multimodal Model-based clinical decision support systems (LMM-CDSS) is characterized by a tension between unprecedented technological acceleration and pronounced systemic fragmentation. As generative AI architectures transition from research proofs-of-concept to real-world clinical deployment at an accelerating pace [1, 2], this advancement is occurring within an expanding, interdisciplinary gap. The necessary integration of computer science, clinical medicine, ethics, and regulatory science [3, 4] has led to pronounced diversity in data modalities, ranging from unstructured Electronic Health Records (EHR) data to intricate multi-sensor time series, and methodological approaches, including Retrieval-Augmented Generation (RAG) and multi-agent paradigms [1, 5]. Although

this diversity is a powerful driver of innovation, it has advanced faster than the formation of shared conceptual frameworks, leaving a significant gap in our understanding of how these disparate components should be integrated into coherent decision-making archetypes that support safe clinical use.

To verify the originality of this study and ensure a non-redundant contribution to the field, a multi-stage preliminary research strategy was executed in February 2026. The preliminary inquiry began with a search for published scoping and systematic reviews at the intersection of clinical CDSS and LMMs. The search scope was constrained to clinical decision-making, deliberately excluding administrative automation, medical training strategies, and patient-facing AI applications. A specific emphasis was placed on the governing attributes of clinical decisions, examining the influence of medical specialties and the inherent complexity of decision-making tasks. The inquiry utilized PubMed (biomedical), IEEE Xplore (technological), and Web of Science (multidisciplinary) databases, supplemented by a targeted audit of the *JBIR Evidence Synthesis* journal and formal registry searches across OSF Registries and PROSPERO. These verification steps confirmed that no existing or ongoing reviews address the specific aim of this study, namely the application of a decision-centric framework aligned with LMM architectural functionalities.

Preliminary research has established that the landscape of LMMs in healthcare is characterized by significant technical velocity. Foundational scoping and systematic reviews have successfully mapped the global distribution of AI applications across broad medical specialties, identifying a high degree of performance heterogeneity across diverse clinical tasks [6, 7]. Re-

cent benchmarks have introduced practical three-tier frameworks to align agent architectures with task complexity, ranging from discrete micro-tasks to multi-agent ecosystems [8]. While these studies provide essential data on architectural alignment and the role of customization pipelines like RAG [9], they typically treat medical specialties as static evaluation categories rather than active governors of decision attributes.

Our protocol proposes moving beyond broad complexity tiers by employing a decision-centric framework that analyzes how the specific nature of a medical specialty dictates decision attributes and, therefore, architectural requirements. Current literature distinguishes between single-agent systems for narrow, vertical domains and multi-agent ecosystems for high-stakes, horizontal problems requiring the synthesis of conflicting evidence [8, 10]. However, a critical gap remains in formalizing how these decisions demand vertical pathophysiological reasoning versus horizontal multidisciplinary integration. While handling long temporal sequences and temporal reasoning are recognized as significant future directions [11], existing reviews often treat clinical tasks as one-off judgments. The current protocol addresses this limitation by distinguishing between static clinical snapshots and longitudinal reasoning trajectories over evolving patient histories.

To maintain precision and clinical relevance, this review is strictly limited to the role of the Digital Practitioner in direct clinical decision-making. This focus distinguishes the current scope from broader foundational works that include healthcare administrative tasks (e.g., billing, scheduling), medical education and licensing exams, or patient-facing self-triage tools [6, 12, 13]. Finally, whereas contemporary research evaluates specialized fine-tuned mod-

els primarily through the lens of training efficiency [9], our approach analyzes how specific clinical case types and specialty-driven constraints force architectural evolution. By centering on these functional decision archetypes, the review provides a more fine-grained taxonomy of procedural complexity than is currently available in the literature.

1.2. Objectives

The primary objective of this scoping review is to map how clinical decision characteristics act as the governor of system design in LMM-driven clinical decision support systems.

Specifically, the review aims to:

- **Characterize the decision space:** Identify and formally represent the principal factors that collectively specify the fundamental properties of a clinical decision space. These include, for example, the level and type of inference required in highly specialized clinical contexts as compared to the degree of integration necessary in cross-specialty or multi-disciplinary systems; the applicable regulatory and normative constraints (e.g., EU AI Act, specialty-specific clinical guidelines); the level of clinical criticality and risk; as well as additional relevant determinants that shape the operational, ethical, and legal boundaries of decision-making processes.
- **Develop a Taxonomy of the Clinical Decision Space:** Construct a taxonomy of decision space archetypes that incorporates the identified characterization. This entails analyzing how the interaction between

these factors collectively determines the structural specifications and constraints of clinical decision-making tasks.

- **Analyze the Architectural Alignment to Clinical Constraints:** Identify how specific clinical decision factors force technical evolutions in system design by mapping the problem/decision Space and the Solution Space. This involves investigating how high-complexity clinical requirements necessitate the integration of advanced architectural paradigms, such as Multi-Agent Systems (MAS) for integrative breadth, temporal modules for longitudinal reasoning, and Advanced Explainability (XAI) or Retrieval-Augmented Generation (RAG) for fact-checking and clinical justification.
- **Evaluate the Determinants of Real-World Clinical Deployment:** Examine how the alignment between decision characteristics and system architecture influences the transition from experimental prototypes to clinical-grade implementation. This involves analyzing how factors such as Advanced Explainability (XAI), fact-checking mechanisms such as RAG, and the management of high-stakes clinical actions drive the development of deployment-ready solutions capable of maintaining logical fidelity and trustworthiness across heterogeneous medical workflows.

1.3. Review questions

The following research questions will guide this scoping review:

Primary Question:

- How do clinical decision characteristics act as the governor for the design and implementation of Large Multimodal Model (LMM)-driven clinical decision support systems?

Sub-questions:

1. **Characterizing the Decision Space:** What are the principal factors (e.g., specialized inference vs. multi-disciplinary integration, regulatory constraints, and clinical risk) that define the operational, ethical, and legal boundaries of a clinical decision space?
2. **Taxonomy of the Decision Space:** How can these identified factors be used to construct a taxonomy of clinical decision archetypes that reflects the structural specifications and constraints of different medical tasks?
3. **Architectural Alignment:** How do specific clinical decision factors force technical evolutions in system design, specifically requiring advanced paradigms such as Multi-Agent Systems, temporal modules, or fact-checking mechanisms like RAG?
4. **Deployment Determinants:** How does the alignment between decision characteristics and system architecture influence the transition from experimental prototypes to clinical-grade implementations that maintain trust and logical fidelity?

2. Inclusion criteria

The inclusion criteria for this scoping review are structured according to the PCC (Population, Concept, Context) framework to ensure direct alignment with the study’s decision-centric objectives.

2.1. Population

This review considers studies involving Digital Practitioners, defined as licensed healthcare professionals (physicians, specialists, nurses, or residents) and multidisciplinary clinical teams utilizing LMM-CDSS within a professional workflow.

- **Inclusion:** Clinicians across all medical specialties and cross-specialty diagnostic teams.
- **Exclusion:** Studies focused on patient-facing applications (self-diagnosis, symptom checkers), on undergraduate medical students (educational tools), or on administrative staff (billing/scheduling).

2.2. Concept

The core concept is the Large Multimodal Model-driven Clinical Decision Support System (LMM-CDSS). The category encompasses Large Multimodal Models (LMMs) and Large Language Models (LLMs) used to process clinical data (text, images, vitals, etc.) for clinical decision support.

- **Inclusion:** The system must produce patient-specific recommendations intended to support a clinical decision (e.g., diagnosis, treatment, prognosis, triage, or discharge planning).
- **Exclusion:**
 - *Non-generative AI:* standalone traditional machine learning (e.g., Random Forest, CNN-only, SVM) or rule-based expert systems lacking a generative transformer-based core.

- *Operational and administrative tasks:* tools used exclusively for administrative automation (e.g., automated scribing, billing, ICD coding) or general medical education and information retrieval without a patient-specific recommendation or decision context.

Terminological note on LMM vs. LLM: This review adopts *Large Multimodal Model* (LMM) as its primary term, following the World Health Organization’s *Guidance on Large Multi-Modal Models* [14], which reflects the architectural shift beyond text-only processing toward native multimodal integration. Because the indexed literature predominantly uses *Large Language Model* (LLM) even for multimodal systems, both terms are retained in all search strings. For inclusion purposes, LLM and LMM designate the same class of generative, transformer-based architectures; the distinction is terminological, not exclusionary.

2.3. Context

The review encompasses all clinical environments where medical decisions are formulated, ranging from traditional infrastructure (Hospitals, Clinics, Emergency Departments) to resource-constrained settings (Rural, Remote, or Mobile-based care). The setting also includes computational environments to accommodate benchmark and validation studies that utilize standardized medical datasets.

- **Inclusion:** This scoping review adopts a global perspective and spans all medical specialties. To trace how generalism and specialization have developed, the review explicitly focuses on two decision frameworks: specialized decisions and cross-specialization decisions.

- **Exclusion:** Non-clinical hospital management (resource allocation, bed management), public health policy modeling without direct practitioner interaction, and medical imaging software that performs only segmentation without a decision-support component.

2.4. *Types of sources and evidence*

This review will consider primary and secondary research published in peer-reviewed journals and conference proceedings.

- **Language:** Only studies published in English will be included.
- **Date Range:** The search will be limited to evidence published from 2024 to the present, reflecting the period in which multimodal architectural paradigms reached sufficient maturity for clinical application.
- **Evidence Types:** Peer-reviewed primary research, conference papers (e.g., NeurIPS, MICCAI), and systematic/scoping reviews for reference harvesting.
- **Grey Literature Policy:** Pre-print repositories such as arXiv are excluded from the primary search for two methodological reasons: pre-prints have not undergone peer review, introducing a quality asymmetry with indexed sources; and this review targets systems with sufficient clinical evaluation, work that predominantly appears in peer-reviewed venues. Grey literature is not categorically excluded; pre-prints or technical reports identified through citation snowballing or cited within included primary sources will be assessed against the inclusion criteria and considered for inclusion. Citation by a peer-reviewed study that

itself meets the inclusion criteria serves as a relevance and credibility signal sufficient to warrant full-text screening, even in the absence of formal peer review of the pre-print itself.

3. Methodology

3.1. Framework and design

Given the characteristics of the LMM-CDS research landscape, this study employs a *scoping review* design to comprehensively synthesize the extant literature, with an emphasis on systematic evidence mapping and the identification of knowledge gaps. To balance methodological rigor with feasibility, a formal critical appraisal of individual study quality is not undertaken, consistent with prevailing conventions for scoping reviews that prioritize evidence mapping and gap detection over the evaluation of intervention effectiveness [15].

This scoping review is conducted in accordance with the Joanna Briggs Institute (JBI) methodological framework [15] and reported following the PRISMA-ScR guidelines to ensure comprehensive transparency [16]. To address the integration of artificial intelligence (AI) as a methodological tool, the review incorporates the PRISMA-trAIce extension which is a discipline-agnostic checklist designed to facilitate the transparent reporting of AI-assisted steps across all stages of evidence synthesis [17]. Within this review, AI assistance is restricted to a single function: reading-aid use during full-text screening. All inclusion and exclusion decisions remain the responsibility of the human reviewers. A single PRISMA-trAIce reporting commitment governs every AI-assisted step in this protocol: at each AI use, the tool or

model name, version, prompting strategy (system and user prompts), and date of use are recorded in the screening database and in the final research paper. The collective application of these standards supports methodological reproducibility while providing a foundation for mapping the current evidence base, delineating research gaps, and informing the direction of future research.

This protocol has been registered on the Open Science Framework (DOI: 10.17605/OSF.IO/K7CDY). Subsequent amendments to the protocol will be recorded as new versions in the same OSF registration to preserve a transparent change history.

3.2. Search strategy

The search strategy will follow the JBI three-step process to ensure a comprehensive, transparent, and reproducible identification of both published and grey literature [15]. In alignment with the Decision-Centric Framework, the strategy is calibrated to capture the intersection of clinical reasoning trajectories and LMM architectural responses.

3.2.1. Initial exploratory search

The initial exploratory search, executed in February 2026, served two complementary functions aligned with JBI step one: confirming the absence of overlapping protocols, and harvesting the high-frequency index terms and keywords that underpin the comprehensive search strings described in Section 3.2.2. This stage functioned as the foundational anchor for defining the boundaries of both the clinical problem space and the technical solution space. An audit was conducted across PubMed, IEEE Xplore, Web of

Science, and the JBI Evidence Synthesis journal. Formal registry searches were simultaneously performed within OSF Registries and PROSPERO to verify that no existing or in-progress protocols address the integration of a decision-centric framework with Large Multimodal Model (LMM) architectural functionalities.

The exploratory search identified several foundational works, particularly regarding Multi-Agent Systems (MAS) in high-criticality healthcare. This preliminary inquiry was instrumental in refining the final search strings through the identification of high-frequency index terms; it also allowed for the precise establishment of inclusion constraints, in particular the exclusion of administrative automation and patient-facing applications, to ensure a dedicated focus on the workflow of the Digital Practitioner.

3.2.2. Comprehensive search strategy

To capture the intersection of clinical reasoning and architectural responses, three primary databases were selected based on their specific indexing strengths and the necessity of cross-validating the Problem-Solution overlap. This triad ensures that the search captures high-impact medical research, current technical innovations, and the interdisciplinary synthesis required for a decision-centric review.

Search terms will be restricted to the title field across all three databases. Restricting to the title is a recognised precision strategy consistent with the scoping review objective of mapping a defined conceptual space: a paper whose primary contribution is LLM-driven clinical decision support will almost invariably reflect that in the title, whereas abstract-only mentions typically indicate incidental framing rather than substantive focus. All records

meeting this scope criterion will be retrieved; the restriction defines the review’s scope, not a statistical sample.

PubMed (National Library of Medicine) serves as the primary reference source for the clinical domain. It offers dedicated indexing of peer-reviewed biomedical literature and MEDLINE records, ensuring that the studies identified are firmly aligned with established clinical practice workflows. Its use is essential for specifying the boundaries of the Digital Practitioner role and for articulating the established concept of a Clinical Decision Support System, which is represented as a medical subheading in MeSH.

Search String (PubMed):

```
((("Decision Support Systems, Clinical"[MeSH Terms] OR "clinical
decision*" [Title]) AND ("large language model*" [Title] OR
"large language*" [Title] OR "LLM" [Title] OR "LLMs" [Title] OR
"large multimodal model*" [Title] OR "LMM" [Title] OR
"LMMs" [Title])) AND ("2024" [Date - Publication] : "2026" [Date -
Publication]) AND (English [Language]))
```

IEEE Xplore (Institute of Electrical and Electronics Engineers) serves as the primary repository for the technical *Solution Space*. Given that LMM architectures, Multi-Agent Systems (MAS), and RAG frameworks often debut in conference proceedings before journal publication, IEEE Xplore is essential for capturing the earliest dissemination of algorithmic innovations.

Search String (IEEE Xplore):

```
("Document Title": "large language model" OR "Document
Title": "large language models" OR "Document Title": "LLM" OR
```

"Document Title":"LMMs" OR "Document Title":"large multimodal model") AND ("Document Title":"clinical decision")

Web of Science (Clarivate) is included to provide *Integrative Breadth*, functioning as a multidisciplinary anchor that captures the social, ethical, and cross-domain implications of AI deployment. Its comprehensive indexing of citation networks allows for mapping the evolution of Trustworthy AI across diverse scientific fields, bridging the gap between theoretical computer science and practical medical ethics. This selection provides a robust mechanism for identifying multidisciplinary studies that may be absent from specialized indices, thereby minimizing selection bias.

Search String (Web of Science Core Collection):

TI=("large language model*" OR "LLM" OR "LMMs" OR "large multimodal model*" OR "LMM") AND TI=("clinical decision*") AND
PY=(2024-2026)

Search strings are adapted to each database's indexing and query syntax. PubMed additionally uses the **Decision Support Systems**, **Clinical MeSH** descriptor, which has no equivalent in IEEE Xplore or Web of Science, where the concept is captured through free-text title terms instead. IEEE Xplore Command Search does not support truncation within title-field queries, so term variants are enumerated literally.

3.2.3. *Supplementary searching*

To ensure maximal coverage and identify relevant literature that may reside outside the search strategy, a backward citation search will be applied to all studies included after Phase 2 full-text screening. Reference lists will be

reviewed to identify seminal foundational works not captured by the primary database search. One level of snowballing will be performed. Candidate references are screened against the same title-level scope criteria applied to the primary search before eligibility assessment, so that the scope of the supplementary search matches the scope of the search that built the corpus. Forward citation searching was omitted given the 2024–2026 search window: papers published within this period have not yet accumulated sufficient citations for forward snowballing to yield meaningful additional coverage.

3.3. Eligibility criteria and study selection

Studies are screened against the eligibility criteria defined in Section 2 (Population, Concept, Context, and Types of Sources and Evidence). At the screening stage, additional disqualifying conditions apply: studies are excluded when their technical content is insufficient to characterize the LMM-CDSS architecture, including opinion pieces, abstract-only conceptual works, and studies that reference only legacy transformer architectures (e.g., BERT) without an LMM-CDSS framing.

The selection process follows a two-phase pipeline. Phase 1 is a broad pre-screening pass based on title and abstract, designed to eliminate clear non-matches; Phase 2 is the definitive screening based on full-text review. Independent human review is the primary mechanism at both phases, with structured reviewer discussion reserved for records where the two reviewers reach different decisions. The final decision at every stage, along with its documented rationale, is recorded in a structured screening database and processed using Python to ensure full methodological transparency and auditability throughout.

Before human screening, records are filtered automatically against the Types of Sources and Evidence criteria in Section 2: language, publication date, source type, and retraction status. These checks are decidable from record metadata alone and involve no reviewer judgment. They apply to records retrieved from the database searches; pre-prints identified through citation snowballing are exempt from the source-type check, in line with the grey literature policy in Section 2. Excluded records are counted and reported in the PRISMA-ScR flow diagram.

Phase 1: Title and Abstract Screening.

Stage 1: Independent Human Screening. Two reviewers independently assess each record by title and abstract, completing the same structured inclusion/exclusion checklist. Each reviewer records one of two decisions: include or exclude, together with a brief written rationale. Records where both reviewers independently agree to exclude are eliminated at this stage, with both rationales logged. Records where both reviewers independently agree to include, and records where the two reviewers reach different decisions, are forwarded to Stage 2.

Stage 2: Reviewer Discussion. For records not resolved in Stage 1, both reviewers meet to discuss the conflicting assessments, each presenting the rationale behind their decision. If discussion produces a consensus, the agreed decision is logged with a joint rationale. If consensus cannot be reached, the record is included. All records included or unresolved at Stage 2, together with all Stage 1 consensus inclusions, proceed to Phase 2.

Phase 2: Full-Text Screening.

All records that passed Phase 1 are retrieved in full. Records for which

the full text cannot be obtained are excluded; these counts and reasons are reported in the PRISMA-ScR flow diagram.

Stage 3: Human Full-Text Review. Two reviewers independently read each full-text record and apply the same inclusion/exclusion checklist. Phase 2 is the definitive screening stage: each reviewer must reach a final include or exclude decision, with a documented rationale logged in the screening database. To support a thorough assessment of complex technical content, reviewers may use an AI reading aid to interrogate specific sections of the full text before reaching a decision; the final decision remains the reviewer’s independent judgment. A single tool and model is fixed in advance and used identically by both reviewers, and is consulted under a predefined set of questions rather than free-form prompting, so that the two reviewers’ AI-assisted sessions remain comparable. The tool issues no include or exclude recommendation. Records excluded at this stage require a specific exclusion reason, reported in the PRISMA-ScR flow diagram. Records where the two reviewers reach different decisions are forwarded to Stage 4. Use of reading-aid AI tools is reported per the PRISMA-trAIce commitment in Section 3.1.

Stage 4: Reviewer Discussion. For records not resolved in Stage 3, both reviewers meet to discuss the conflicting assessments, each presenting the rationale behind their full-text evaluation. If discussion produces a consensus, the agreed decision is logged with a joint rationale. If consensus cannot be reached, the record is included.

3.4. Evidence charting and data management

The extraction process will utilize a standardized table, iteratively refined to capture the intersection of clinical decision archetypes and LMM archi-

tectural responses. Data charting will be performed by a single reviewer using the standardized table, with all entries traceable to the source paper to allow subsequent verification by the research team. The final selection of studies will be archived in a version-controlled repository, stored primarily in .csv format to ensure long-term interoperability and immediate accessibility via Python-based analytical frameworks. This dataset functions as the formal evidentiary basis for the review, maintaining a granular audit trail that includes the rationale for inclusion or exclusion at every stage.

The extraction instrument will systematically capture variables across three primary dimensions: *(i) Study Characteristics* (authors, year, clinical specialty); *(ii) The Problem Space* (decision criticality, reasoning trajectory, and workflow integration); and *(iii) The Solution Space* (LMM architecture, use of Multi-Agent Systems, RAG implementation, and XAI functionality).

3.5. Python-driven synthesis and visualization

Following the extraction phase, the raw data will be ingested into a Pandas dataframe for cleaning and synthesis. This computational approach allows for the automated validation of data consistency and the calculation of inter-rater reliability metrics between the two independent human reviewers at each screening phase, alongside descriptive statistics on the proportion and direction of disagreements resolved through reviewer discussion.

The final data synthesis will be visualized using Python libraries (e.g., Matplotlib, Seaborn) to produce evidence maps. These visualizations will include a PRISMA-ScR flow diagram detailing the study selection process and multidimensional charts mapping clinical constraints to architectural responses. This programmatic pipeline ensures that the transition from raw

search results to the final scoping review is reproducible and aligned with the project’s decision-centric framework.

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